

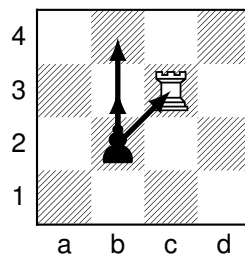
Chess

IMO Training Level 2

20/07/2021

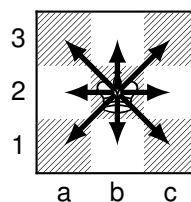
There are 6 types of pieces in chess: pawn, knight, rook, bishop, queen and king, each with its own characteristic way of moving on the board, and these serve as a very good inspiration for math Olympiad questions. Let's study the different types of pieces (and problems related to them) one by one.

1 Pawn



A pawn can move one or two step forward (from the point of view of that player) if there is not obstacles, and if a pawn want to capture another piece, it must do so by moving diagonally forward. Due to their complicated rules of movement, pawns seldom appear in math Olympiad problems.

2 King



The king has quite a low mobility, considering that it is the piece that determines the result of the game. The rule is simple - it attacks any cell sharing at least one vertex with the cell the king itself is on.

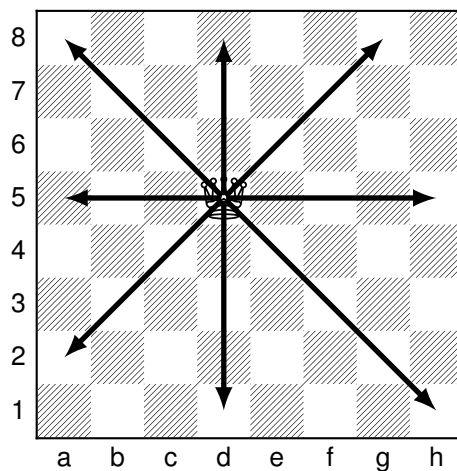
1. [1] How many kings can be placed on an $n \times n$ chessboard such that no two of them attack each other?
2. [1] Show that in a king's tour (in which a king visits each of 64 squares exactly once, then return to its starting point), the number of diagonal moves must be an even number.

3. [3] (IMO 2010 C3) 2500 chess kings have to be placed on a 100×100 chessboard such that :

- (i) no king can capture any other one
- (ii) each row and each column contains exactly 25 kings

Find the number of such arrangements. (Two arrangements differing by rotation or symmetry are supposed to be different)

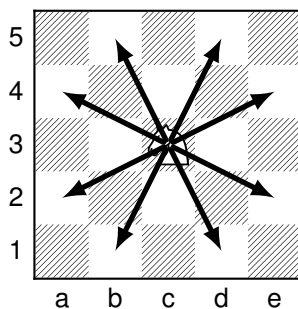
3 Queen



Queen is the most powerful piece in chess: it can move in any straight line and any diagonal for any length, as long as there are no obstacles.

- 4. [1] At least how many queens should be placed on a 6×6 chessboard, so that for each square, either a queen is on it or it is attacked by a queen?
- 5. [2] (n queens problem) At most how many queens can be placed on an 8×8 chessboard, so that no two of them attack each other?

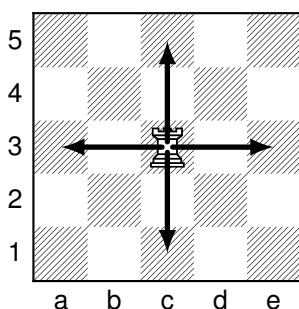
4 Knight



The knight is probably the piece with the most peculiar rules for movement: it moves to a square which is at a distance of $\sqrt{5}$ away from it (considering the distance between centres of squares). Because of this, knight is one of the pieces that appears most frequently in math Olympiad.

6. [0] Convince yourself that a knight always move from a black square to a white square, or vice versa.
7. [1.5] It is a famous fact that there exists a knight's tour on an 8×8 chessboard. Try to construct one.
8. [1] Does there exist a knight's tour on a 4×4 chessboard?
9. [2.5] For what n does there exist a knight's tour on a $4 \times n$ chessboard?
10. [1] At most how many knights can we place on an 8×8 chessboard so that no two of them attack each other?
11. [1.5] n knights are placed on a chessboard, so that there exists a positive integer k such that each knight attacks exactly k other knights. Prove that n is even.
12. [2] (Caucasus MO) On an 8×8 chessboard, $n > 6$ Knights are placed, so that for any 6 Knights among them, there are two Knights that attack each other. Find the greatest possible value of n .
13. [3] (IMO 2018 P4) A site is any point (x, y) in the plane such that x and y are both positive integers less than or equal to 20. Initially, each of the 400 sites is unoccupied. Amy and Ben take turns placing stones with Amy going first. On her turn, Amy places a new red stone on an unoccupied site such that the distance between any two sites occupied by red stones is not equal to $\sqrt{5}$. On his turn, Ben places a new blue stone on any unoccupied site. (A site occupied by a blue stone is allowed to be at any distance from any other occupied site.) They stop as soon as a player cannot place a stone. Find the greatest K such that Amy can ensure that she places at least K red stones, no matter how Ben places his blue stones.

5 Rook

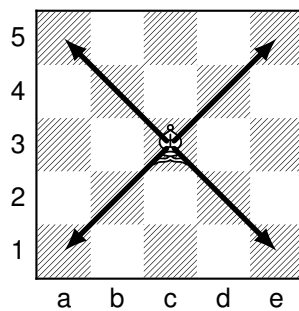


Rooks are very simple, they move vertically or horizontally, and they move as far as they want as long as no obstacles come into their way.

14. [1] How many ways are there to place 8 rooks in an 8×8 chessboard so that each rook is on a white square, and no two rooks attack each other?
15. [2] (ToT 2005) Originally, every square of 8×8 chessboard contains a rook. One by one, rooks which attack an odd number of others are removed. Find the maximal number of rooks that can be removed. (A rook attacks another rook if they are on the same row or column and there are no other rooks between them.)

16. [2] (St. Petersburg 2016) On a 300×300 board, several rooks are placed so that they together attack the whole board, and each rook is attacked by no more than 1 other rook. Find the smallest k , such that, no matter how the rooks are placed, we can always find a $k \times k$ square which contains at least one rook.
17. [3] (IMO 2014 C3) Let $n \geq 2$ be an integer. Consider an $n \times n$ chessboard consisting of n^2 unit squares. A configuration of n rooks on this board is *peaceful* if every row and every column contains exactly one rook. Find the greatest positive integer k such that, for each peaceful configuration of n rooks, there is a $k \times k$ square which does not contain a rook on any of its k^2 squares.

6 Bishop



Bishops move in a way very similar to rooks, except that they move in diagonals. It is an obvious, while sometimes useful observation that bishops always stay in white (or black) squares.

18. [3] (HMIC 2020 P2) Some bishops and knights are placed on an infinite chessboard, where each square has side length 1 unit. Suppose that the following conditions hold: For each bishop, there exists a knight on the same diagonal as that bishop (there may be another piece between the bishop and the knight). For each knight, there exists a bishop that is exactly $\sqrt{5}$ units away from it. If any piece is removed from the board, then at least one of the above conditions is no longer satisfied. If n is the total number of pieces on the board, find all possible values of n .